

Enterprise Smart Supply Chain & Logistics Management System

Business Requirements Specification — Final Baseline

Document Version: 1.0

Status: Approved Baseline

System Type: Multi-Company Enterprise Supply Chain & Logistics Platform

1. Business Objective

The organization requires a centralized enterprise system to manage its end-to-end supply-chain and logistics operations across multiple companies.

The system shall provide a unified platform for:

- Organization management
- Product catalog management
- Supplier and procurement management
- Warehouse management
- Inventory management
- Sales order management
- Payment tracking
- Order fulfillment
- Shipment and delivery tracking
- Returns and reverse logistics
- Refund processing
- Notifications and operational alerts
- Auditing and operational traceability
- Reporting and analytics

The system must preserve historical business information and provide sufficient traceability to determine:

Who performed an operation, what happened, when it happened, where it happened, and which business transaction caused it.

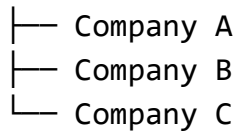
2. Enterprise Structure

BR-ORG-001 — Enterprise

The system shall support a top-level **Enterprise** containing multiple companies.

An Enterprise may contain:

Enterprise



BR-ORG-002 — Company

Each Company shall operate as an independent business entity within the Enterprise.

A Company shall have:

- branches
- departments
- employees
- customers
- suppliers
- products it sells
- warehouses it is authorized to use
- sales transactions
- procurement transactions

BR-ORG-003 — Branches

A Company may operate multiple Branches.

Each Branch belongs to exactly one Company.

BR-ORG-004 — Departments

A Company may have multiple Departments.

Departments are organizational units and are independent from the physical warehouse structure.

BR-ORG-005 — Employees

Employees belong to a Company and may be associated with organizational departments and business roles.

An employee may perform operational activities such as:

- sales
- warehouse operations
- procurement
- delivery
- administration
- finance

BR-ORG-006 — Multiple Departments

An employee may participate in more than one Department where required by the business.

BR-ORG-007 — Business Roles

An employee may have multiple business roles.

Examples:

- Sales Employee

- Warehouse Employee
- Procurement Officer
- Delivery Manager
- Finance Employee
- Administrator

Business roles are separate from technical database permissions.

3. Address Management

BR-ADDR-001

The enterprise shall maintain reusable physical addresses for business entities such as:

- Companies
- Branches
- Suppliers
- Warehouses

BR-ADDR-002

Customer addresses shall support customer-specific information such as:

- address type
- default address
- active/inactive state

BR-ADDR-003 — Historical Order Address

When a customer places an order, the shipping/billing address used by the order shall remain historically preserved.

Changing the customer's current address later must not change the historical address of an existing order.

4. Warehouse Management

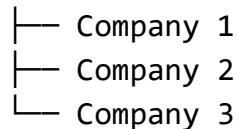
BR-WH-001 — Shared Warehouses

A Warehouse is an enterprise physical resource and is **not inherently owned by a single Company**.

A Warehouse may serve multiple Companies.

Example:

Warehouse A



BR-WH-002 — Company Warehouse Access

A Company must explicitly be authorized to use a Warehouse.

The system shall maintain:

- access type
- start date
- end date
- status

BR-WH-003 — Warehouse Zones

A Warehouse may contain multiple Zones.

Examples:

- Receiving
- Storage
- Picking
- Packing

- Damaged
- Returns
- Repair

BR-WH-004 — Storage Bins

Each Zone may contain multiple Storage Bins.

A Storage Bin belongs to exactly one Warehouse and one Zone.

BR-WH-005

The system shall prevent a Company from storing or moving inventory in a Warehouse to which the Company has no valid access.

5. Product Catalog

BR-PROD-001 — Global Product Catalog

The enterprise shall maintain a centralized Product Catalog.

Products are not duplicated for every Company.

BR-PROD-002 — Brand

Products may belong to a Brand.

A Brand may contain multiple Products.

BR-PROD-003 — Product Variant / SKU

A Product may have multiple sellable variants.

Examples:

Product: T-Shirt

Variant 1 → Black / Large

Variant 2 → Black / Medium

Variant 3 → White / Large

Each Variant shall represent a distinct sellable SKU.

BR-PROD-004 — SKU

Each Product Variant shall have a unique SKU.

A barcode may optionally be assigned to a Variant.

BR-PROD-005 — Categories

Products may belong to multiple Categories.

A Category may contain subcategories.

Example:

```
Electronics
├── Computers
│   ├── Laptops
│   └── Desktops
└── Accessories
```

BR-PROD-006 — Company Product Configuration

A Company may sell selected Product Variants.

Each Company may define its own:

- selling price
- cost
- reorder level
- reorder quantity
- minimum stock
- maximum stock
- active/inactive status

BR-PROD-007

A Company shall not sell a Product Variant unless that Variant has been enabled for the Company.

6. Currency & Tax

BR-FIN-001 — Currency

The system shall support multiple currencies.

Transactions shall retain the currency used at transaction time.

BR-FIN-002 — Historical Financial Values

Changing a product price, supplier cost, or currency configuration shall not alter historical transactions.

BR-TAX-001

The system shall maintain tax rates with effective dates.

BR-TAX-002

Historical transactions shall preserve the applicable tax information used when the transaction occurred.

7. Supplier Management

BR-SUP-001

The enterprise shall maintain a centralized Supplier directory.

BR-SUP-002 — Company-Supplier Relationship

A Company may work with selected Suppliers.

A Supplier may work with multiple Companies.

BR-SUP-003

The Company-Supplier relationship shall support:

- supplier code
- payment terms
- credit limit
- relationship status
- start date
- end date

BR-SUP-004 — Supplier Products

A Supplier may provide multiple Product Variants.

The system shall maintain supplier-specific:

- Supplier SKU
- Unit Cost
- Minimum Order Quantity
- Lead Time
- Preferred supplier status

8. Procurement

BR-PO-001

A Company shall be able to create Purchase Orders for authorized Suppliers.

BR-PO-002

A Purchase Order shall identify:

- Company
- Supplier
- intended receiving Warehouse
- order date
- expected date
- status
- financial totals
- creator

BR-PO-003 — Purchase Order Items

Each Purchase Order may contain multiple Product Variants.

Each item shall record:

- ordered quantity
- agreed unit cost

BR-PO-004 — Historical Supplier Cost

The Purchase Order shall preserve the agreed cost at the time of purchase.

Future changes to the Supplier's current cost must not modify existing Purchase Orders.

BR-PO-005 — Multiple Receipts

A Purchase Order may be received through multiple Goods Receipts.

Example:

Ordered = 1,000

Receipt #1 = 400

Receipt #2 = 350

Receipt #3 = 250

BR-PO-006

The total quantity received for a Purchase Order Item shall never exceed the ordered quantity.

BR-PO-007

The actual receiving Warehouse may differ from the Warehouse originally specified on the Purchase Order.

BR-PO-008 — PO Status

Purchase Orders shall support:

DRAFT

SUBMITTED

APPROVED
PARTIALLY_RECEIVED
FULLY_RECEIVED
CANCELLED
REJECTED

Valid transitions shall be controlled by business workflow.

BR-PO-009 — Status History

Every Purchase Order status transition shall be traceable.

The system shall record:

- previous status
- new status
- employee
- timestamp
- reason when applicable

9. Inventory

BR-INV-001

Inventory shall be tracked by:

- Company
- Warehouse
- Storage Bin
- Product Variant

BR-INV-002

The same Product Variant may exist:

- in multiple Warehouses

- in multiple Bins
- for multiple Companies

BR-INV-003 — Inventory Ownership

Inventory is logically owned by the Company even when physically stored in a shared Warehouse.

BR-INV-004 — Inventory Quantities

The system shall track:

- On Hand
- Reserved
- Damaged

Available quantity shall be derived as:

$\text{Available} = \text{OnHand} - \text{Reserved} - \text{Damaged}$

BR-INV-005

Inventory quantities shall never become negative.

BR-INV-006

Reserved quantity shall never exceed On-Hand quantity.

BR-INV-007

Damaged quantity shall not exceed On-Hand quantity.

10. Inventory Transactions

BR-INV-008

Every physical inventory movement shall be traceable.

Supported movement types shall include:

PURCHASE_RECEIPT

SALE

RETURN

TRANSFER_IN

TRANSFER_OUT

ADJUSTMENT

DAMAGE

BR-INV-009

Every Inventory Transaction shall identify:

- affected inventory
- quantity
- transaction type
- reference to the originating business operation where applicable
- employee
- date/time
- reason where applicable

BR-INV-010

Inventory Reservation is a logical commitment and shall not be treated as a physical stock movement.

11. Inventory Reservation

BR-RES-001

The system shall support reserving inventory for Sales Orders.

BR-RES-002

A reservation shall reference the relevant Order Item and Inventory location.

BR-RES-003

The system shall prevent concurrent orders from reserving more inventory than is available.

BR-RES-004

Reservations shall support lifecycle operations such as:

- creation
- fulfillment
- release
- cancellation

12. Inventory Transfer

BR-TR-001

The system shall support transferring inventory between locations.

Transfers may occur:

- between Warehouses
- between Bins
- between authorized Company/Warehouse combinations according to business rules

BR-TR-002

A Transfer shall contain one or more Product Variants.

BR-TR-003

The system shall track:

- requested quantity
- shipped quantity
- received quantity

BR-TR-004

The following must always hold:

Shipped $\leq$ Requested

Received $\leq$ Shipped

BR-TR-005

Inventory shall leave the source location when the transfer is shipped.

BR-TR-006

Inventory shall enter the destination location when the transfer is received.

BR-TR-007 — Transfer Lifecycle

REQUESTED

→ APPROVED

→ RESERVED

→ SHIPPED

→ IN_TRANSIT

→ RECEIVED

→ COMPLETED

A Transfer may be rejected where permitted by business workflow.

13. Customer Management

BR-CUST-001

Customers shall exist at Enterprise level.

A Customer may interact with multiple Companies.

BR-CUST-002

A Company must establish a valid Company-Customer relationship before processing orders for that customer.

BR-CUST-003

The system shall maintain customer contact and status information.

14. Sales Orders

BR-SALES-001

A Sales Order belongs to exactly one Company and one Customer.

BR-SALES-002

The Company must have a valid business relationship with the Customer.

BR-SALES-003

A Sales Order may contain multiple Order Items.

BR-SALES-004 — Historical Selling Price

Each Order Item shall preserve the actual selling price used at the time of sale.

Later changes to the Company's Product price must not change existing orders.

BR-SALES-005

The Order shall preserve transaction-level:

- subtotal
- discount
- tax
- shipping
- total
- currency

BR-SALES-006

Historical financial values must remain unchanged after the transaction is completed.

15. Sales Order Lifecycle

Valid operational states shall include:

DRAFT

→ PENDING_PAYMENT

→ PAID

→ ALLOCATED

→ PICKING

→ PACKED

→ SHIPPED

→ DELIVERED

Alternative flows include:

PENDING_PAYMENT → CANCELLED

and return operations are handled through the dedicated Returns process.

16. Order Status History

Every Sales Order status change shall be traceable through a history record containing:

- old status
- new status
- employee
- timestamp
- reason

17. Payments

BR-PAY-001

An Order may have multiple payment attempts.

Example:

Attempt 1 → FAILED

Attempt 2 → FAILED

Attempt 3 → SUCCESS

BR-PAY-002

Successful payments shall not exceed the Order total.

BR-PAY-003

Successful payment transactions shall have a transaction reference.

BR-PAY-004

Payment status shall distinguish successful and failed attempts.

18. Fulfillment

BR-FUL-001

A Sales Order may be fulfilled from multiple Warehouses.

BR-FUL-002

Each Fulfillment shall identify the Warehouse responsible for fulfilling it.

BR-FUL-003

The Warehouse must be authorized for the Company owning the Order.

BR-FUL-004 — Split Fulfillment

An Order Item may be split across multiple Fulfillments.

Example:

Ordered = 100

Warehouse A = 60

Warehouse B = 40

BR-FUL-005

The total fulfilled quantity for an Order Item shall never exceed its ordered quantity.

BR-FUL-006 — Fulfillment Lifecycle

PENDING

→ ALLOCATED

→ PICKING

→ PACKED

→ READY_TO_SHIP

→ SHIPPED

→ COMPLETED

Cancellation may occur according to business rules.

19. Shipment

BR-SHIP-001

A Fulfillment may generate multiple Shipments.

BR-SHIP-002

Each Shipment shall be handled by a Carrier.

BR-SHIP-003

A Shipment may have a tracking number.

BR-SHIP-004 — Partial Shipment

A Fulfillment Item may be shipped across multiple Shipments.

Example:

Fulfillment = 100

Shipment 1 = 40

Shipment 2 = 60

BR-SHIP-005

The total shipped quantity shall never exceed the Fulfillment quantity.

BR-SHIP-006

A Shipment belongs to one Fulfillment; therefore its fulfillment Warehouse is the operational source of the shipment.

20. Shipment Tracking

The system shall maintain tracking events such as:

CREATED
PICKED_UP
IN_TRANSIT
ARRIVED
OUT_FOR_DELIVERY
DELIVERED

Additional exception states may include:

FAILED
RETURNED

Each event shall record:

- status
- location where available
- event time
- description

21. Shipment Lifecycle

Typical lifecycle:

CREATED
→ READY
→ PICKED_UP
→ IN_TRANSIT
→ ARRIVED

- OUT_FOR_DELIVERY
- DELIVERED

The system shall preserve shipment history rather than overwriting historical events.

22. Delivery

BR-DEL-001

A Shipment shall not be considered delivered until an appropriate delivery event has occurred.

BR-DEL-002

DeliveredAt shall not precede ShippedAt.

BR-DEL-003

The system shall support measuring:

- shipping duration
- transit duration
- delivery delay
- carrier performance
- on-time delivery

23. Returns

BR-RET-001

A Return Request may be created for a delivered Sales Order according to the company's return policy.

BR-RET-002

A Return Request may contain multiple returned Order Items.

BR-RET-003

A customer may return only quantities that were actually purchased.

BR-RET-004

Multiple Return Requests may exist for the same Order where allowed, but cumulative approved/received quantities shall never exceed the purchased quantity.

24. Return Lifecycle

REQUESTED

→ APPROVED

→ IN_TRANSIT

→ RECEIVED

→ INSPECTING

→ DISPOSITIONED

→ REFUND_PENDING

→ REFUNDED

→ COMPLETED

Alternative:

REQUESTED → REJECTED

25. Return Receipt

Returned products may arrive through one or more Return Receipts.

Each Receipt shall identify:

- Warehouse
- receiver
- receipt time
- status

The receiving Warehouse must be authorized for the relevant Company.

26. Return Inspection

Every received return shall be subject to inspection according to business policy.

Inspection results may include:

RESELLABLE

DAMAGED

DEFECTIVE

MISSING_PARTS

REPAIR_REQUIRED

SCRAP

The inspection result represents the actual physical inspection, not merely the customer's reported condition.

27. Return Inspection Quantities

A single return may have different conditions.

Example:

Returned = 10

6 → RESELLABLE
2 → REPAIR_REQUIRED
1 → DAMAGED
1 → SCRAP

Therefore inspection must support quantities per result.

28. Return Disposition

The business shall determine what happens to inspected products.

Possible dispositions:

RESTOCK
REPAIR
SCRAP
DAMAGED
RETURN_TO_SUPPLIER

29. Return → Inventory

A successfully inspected and restockable item shall increase available inventory.

Damaged products shall not automatically become saleable inventory.

Scrapped products shall not be added to saleable inventory.

All physical inventory changes caused by returns shall be recorded in the inventory movement history.

30. Refunds

BR-REF-001

A Refund is a financial operation separate from the physical Return process.

BR-REF-002

A Return may be approved or received before the Refund is processed.

BR-REF-003

The total successful refunds shall not exceed the amount successfully paid by the customer.

BR-REF-004

The Refund shall not exceed the eligible returned value according to the company's return/refund policy.

BR-REF-005

Successful refunds shall have a transaction reference.

31. Notifications & Alerts

The system shall support operational notifications.

Examples:

LOW_STOCK

DELAYED_SHIPMENT

PENDING_RETURN

OVERDUE_PURCHASE_ORDER

BR-NOT-001

Notifications may be directed to individual Employees.

BR-NOT-002

The system shall track whether a notification was read.

BR-NOT-003

Notifications may have:

- priority
- creation time
- expiration time
- status

32. Alert Rules

The system shall support configurable alert rules.

Examples:

Stock < Reorder Level
Shipment delayed > X hours
PO overdue > X days
Return pending > X days

Rules may be:

- Enterprise-wide
- Company-specific

33. Alert Deduplication

The system shall avoid repeatedly generating identical alerts for the same unresolved business condition.

The system shall maintain the lifecycle of an alert:

ACTIVE

→ RESOLVED

with detection timestamps.

34. Auditing

The system shall provide auditability for important business changes.

Audit information shall support:

- what changed
- old value
- new value
- record affected
- employee/user
- timestamp
- operation type

35. Error Logging

Application/database errors shall be recorded with sufficient diagnostic information, including where applicable:

- error number
- severity
- state
- procedure
- line

- message
- user
- timestamp

36. Security

The system shall distinguish between:

Business Roles

Sales
Warehouse
Procurement
Finance
Delivery
Administrator

and:

Technical Database Roles

db_sales
db_warehouse
db_procurement
db_finance
db_reporting
db_auditor

Business roles define what a person does.

Database roles define what database permissions a technical account receives.

37. Multi-Company Data Isolation

Users operating for one Company shall not automatically have access to another Company's operational data.

The system shall support company-level data isolation.

Where appropriate, Row-Level Security shall be used to enforce this at database level.

38. Historical Integrity

The system must preserve historical truth.

Changing current:

- Product Price
- Supplier Cost
- Customer Address
- Tax Rate
- Company Configuration
- Warehouse relationship

must not silently modify completed historical transactions.

39. Concurrency Requirements

The system shall support concurrent operations.

Particularly sensitive operations include:

Inventory Reservation

Two customers must not successfully reserve the same last available unit.

Purchase Receipt

Two users must not receive more quantity than was ordered.

Fulfillment Allocation

Two fulfillment operations must not allocate more quantity than the Order requires.

Returns

Two return requests must not collectively return more quantity than purchased.

Refunds

Concurrent refund operations must not exceed the eligible refundable amount.

These operations shall use transactional control and appropriate locking/isolation mechanisms.

40. Transaction Requirements

Critical business operations shall be atomic.

For example:

Purchase Receipt

Validate

↓

Lock

↓

Receive

↓

Update Inventory

↓

Create Inventory Transaction

↓

Update PO Status

↓

Commit

If any step fails:

ROLLBACK

41. Traceability Requirement

For important business operations, the system should be able to answer:

Who did it?

When?

What changed?

Which entity was affected?

Which business transaction caused it?

Examples:

Purchase Order

Sales Order

Payment

Shipment

Return

Inventory Movement

Refund

42. Reporting Requirements

The system shall support reporting for at least:

Sales

- sales by Company
- sales by Product
- sales by Category
- sales by Customer
- sales over time

Inventory

- current stock
- available stock

- reserved stock
- damaged stock
- low-stock products
- stock turnover
- inventory reconciliation

Procurement

- supplier spending
- supplier lead time
- purchase fulfillment
- outstanding POs
- overdue POs

Logistics

- shipments by Carrier
- average delivery time
- delayed shipments
- on-time delivery
- warehouse fulfillment performance

Returns

- return rate
- return reasons
- inspection outcomes
- disposition outcomes
- refund values

Finance

- payments
- failed payment attempts
- refunds
- revenue-related metrics

43. Operational Automation Requirements

The system shall support scheduled operational tasks such as:

Low-stock detection
Delayed shipment detection
Pending return detection
Overdue PO detection
Audit/log maintenance
Index maintenance
Database backups
Reporting refresh/maintenance

The implementation may use SQL Server Agent.

44. Performance Requirements

The database shall be designed to support:

- high-volume transactional data
- frequent inventory updates
- concurrent orders
- large historical transaction tables
- reporting workloads

The implementation shall support performance analysis through:

- indexes
- execution plans
- logical reads
- CPU/time measurement
- query tuning
- partitioning where justified

45. Data Integrity Requirements

The system shall enforce integrity at the database/business-logic level where practical.

Examples:

Invalid Company/Warehouse relationship

Invalid Company/Customer relationship

Negative inventory

Over-receiving

Over-fulfillment

Over-shipment

Over-return

Over-refund

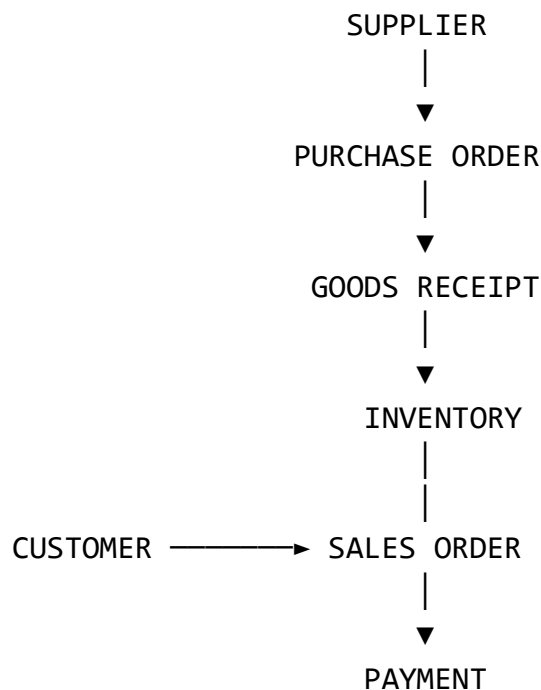
Invalid transfer quantities

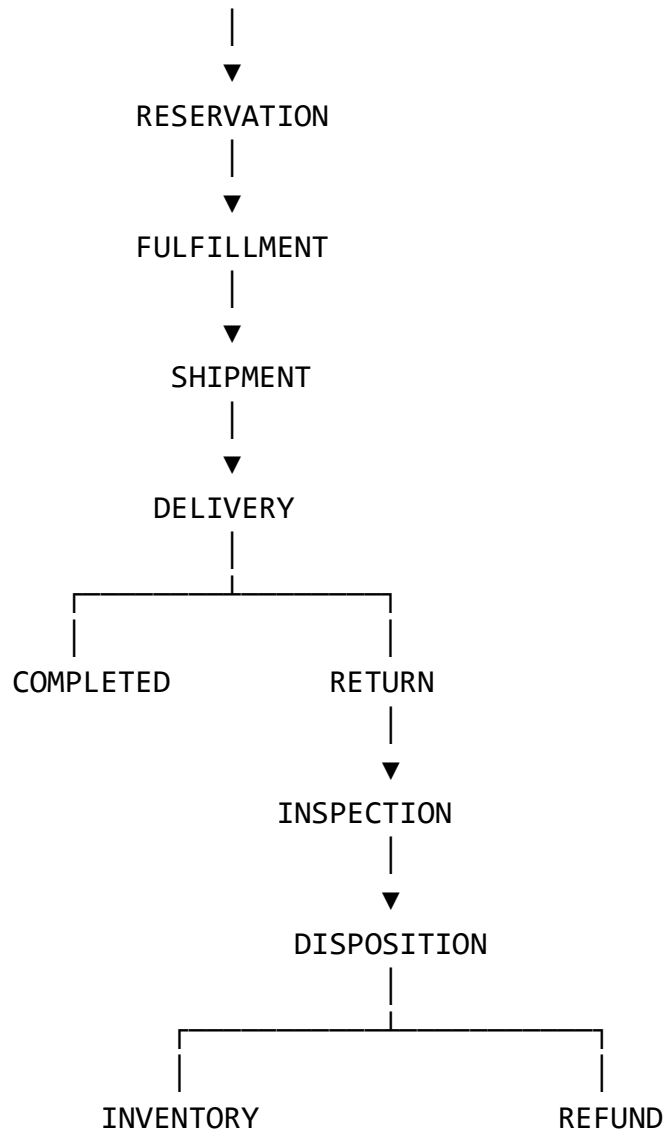
Invalid status transitions

The application must not be the only layer protecting critical business rules.

46. Final End-to-End Business Flow

وده أهم جزء في الوثيقة كلها:





🔗 أهم حاجة: إيه اللي يعتبر Requirement وإيه اللي إحنا كـ Engineers قررناه؟

علشان لما ترجع للوثيقة بعد شهر ما يحصلش خطأ.

Business Requirements

دي اللي فوق، مثل:

Company ممكن يخدم أكثر من Warehouse.

Warehouse ممكن يتنفذ من أكثر من Order.

OrderItem يمكن يتقسم

يمكن يوصل على دفعات PO

العميل ممكن يعمل Return.

لا يتجاوز المدفوع Refund

Technical Design Decisions

أما حاجات زي:

Composite FK
3NF
Schema separation
Filtered Index
RLS
Stored Procedure
Trigger
SQL Server Agent
Partitioning
Covering Index

Business Requirements. دي مش

دي قراراتنا إحنا علشان نحقق الـ Business Requirements.

وده فصل مهم جدًا في المشروع.

Baseline اللي هنشغل عليه

اعتبر الوثيقة دي من دلوقتي:

Business Requirements Baseline v1.0

يعني لما نبدأ SQL، كل جدول وكل Constraint وكل Procedure لازم نقدر نجاب:

"الـ Business Requirement اللي خلاني أعمل ده إيه؟"